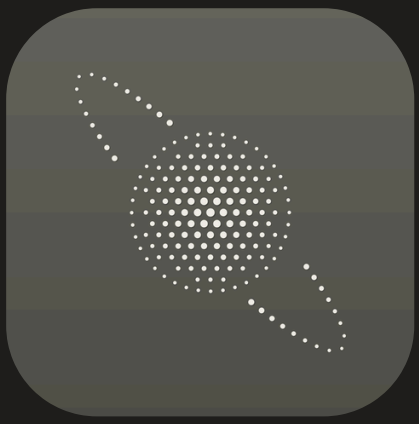


# S Æ T E R N E S D Æ G

ONE OSCILLATOR, FED BACK

# One oscillator, fed back

Sæternesdæg is one oscillator run through a **shaper**, a **pulser** and an **XOR**, with the end of the chain fed back to its start. Four pads move the circuit; a second oscillator sits beside it, and slow modulators — chaos, two sample & holds, a sequencer, an envelope — move the rest.

The same circuit comes three ways: **ORIGIN** (the app's circuit, played on the pads), **SYNTH** (eight voices from its parts, played from a keyboard) and **MODULAR** (every inner connection cut: you patch it with cables).

**Too much to read? Skip it.** Open PRESET, then Setting and press RAND ALL. Drag the four pads. Hold a preset number to keep what you hear.

**In your DAW**

The app, the AU (AUv3) and the VST3 are the same instrument. In your DAW you'll find **Sæternesdæg** under *k.odk*. In ORIGIN it plays on its own, a drone; MIDI notes set its pitch. SYNTH plays from MIDI notes.

## Installing

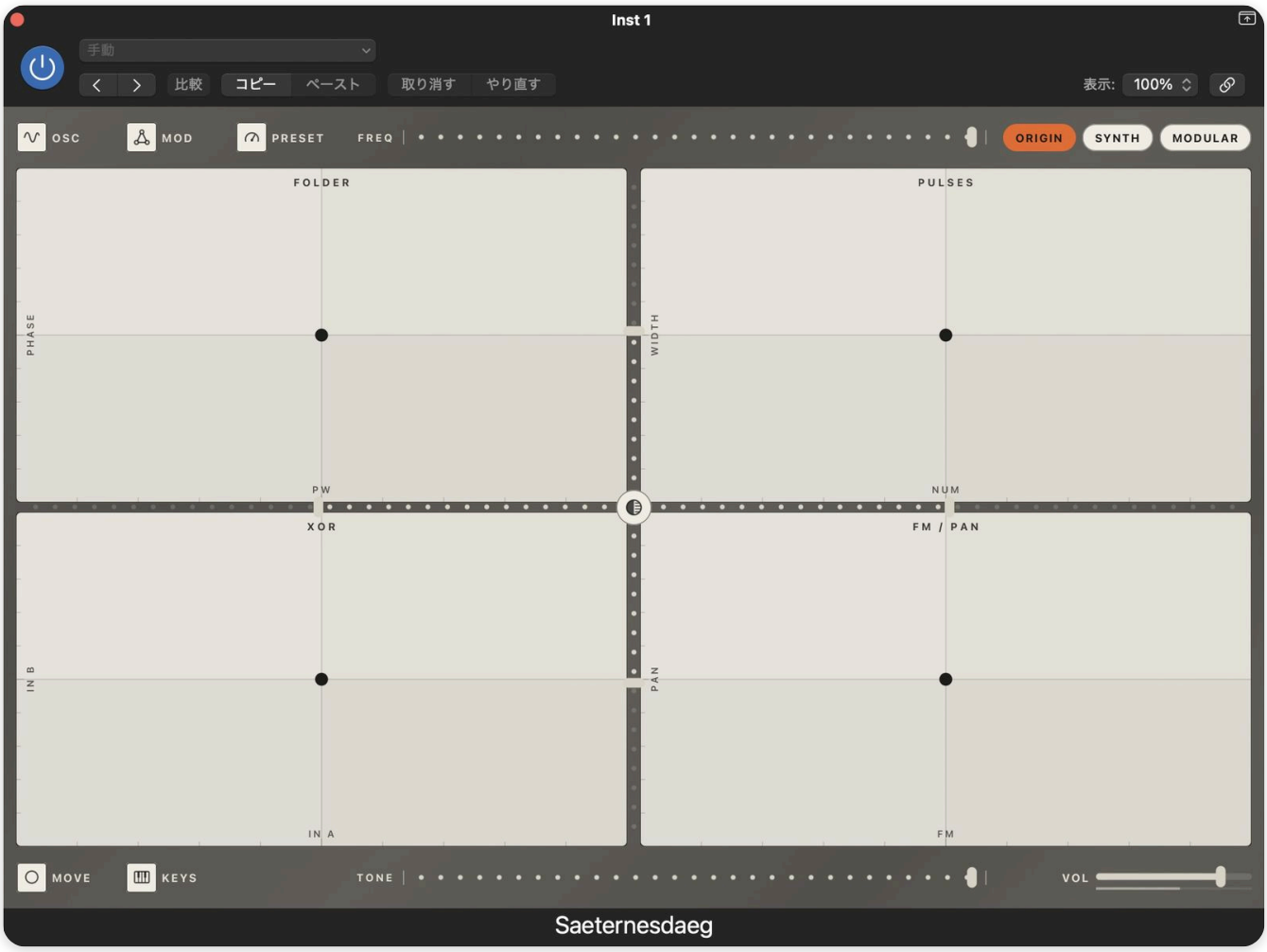
1. Open **Sæternesdæg.pkg** and follow the installer. You can choose: **Standalone + AU**, and/or **VST3**.
2. Open **Sæternesdæg** once from Applications — this registers the AUv3.
3. In your DAW, rescan plug-ins. Sæternesdæg is under **k.odk**.

The camera is optional: the first time the moon is turned on, macOS asks for permission.

| | |
|---|---|
| Standalone + AU | The Sæternesdæg app in /Applications, with its Audio Unit (AUv3) inside: Logic, GarageBand, Live and other AU hosts list it under *k.odk*. The camera works in both. |
| VST3 | In /Library/Audio/Plug-Ins/VST3 — Live, Bitwig, Reaper, Cubase … |
| The app | Plays on its own, through the Mac's output. If that output goes away (headphones unplugged, a Bluetooth device taken by another device), it opens its output again by itself. |

Requires macOS 12 or later · Apple silicon and Intel.

# The two bars




| | |
|---|---|
| OSC · MOD · PRESET | The three panels: the oscillators and the sound path, the modulators, and Setting (the output, the camera, the presets). They open over the screen; **Done** or Esc closes them. |
| ORIGIN · SYNTH · MODULAR | The mode. A change crossfades; each mode keeps its own sound and its own presets. |
| FREQ · TONE | The long dotted sliders at the top and the bottom. In ORIGIN: the oscillator's frequency and the output tone. In SYNTH: the filter's CUT and RES. In MODULAR: TZ OSC 1's frequency and the 3-pole's CUT. |
| MOVE | The four pads recorded together (12 s). Tap: record, loop, stop. Double tap: clear. |
| KEYS | The keyboard along the bottom (always out in SYNTH). **Z −** and **+ X** move it an octave; the computer's keys A W S E D F T G Y H U J K play it. |
| VOL | The output volume, with the level under it. Double-click: 80 %. |

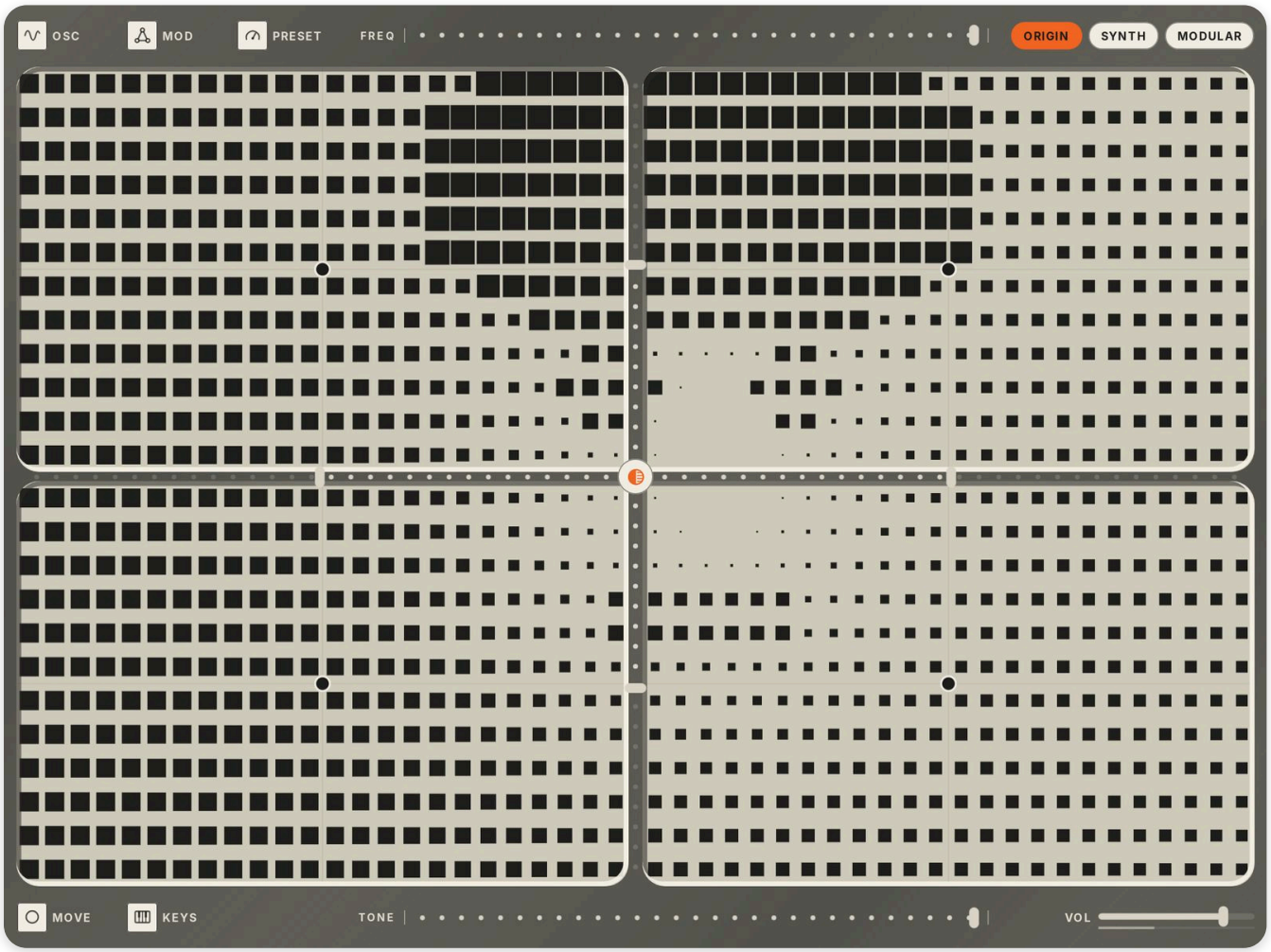
# Four pads, a cross, the moon

Each pad moves two of the circuit's knobs: its name on top, X along the bottom, Y up the left side.

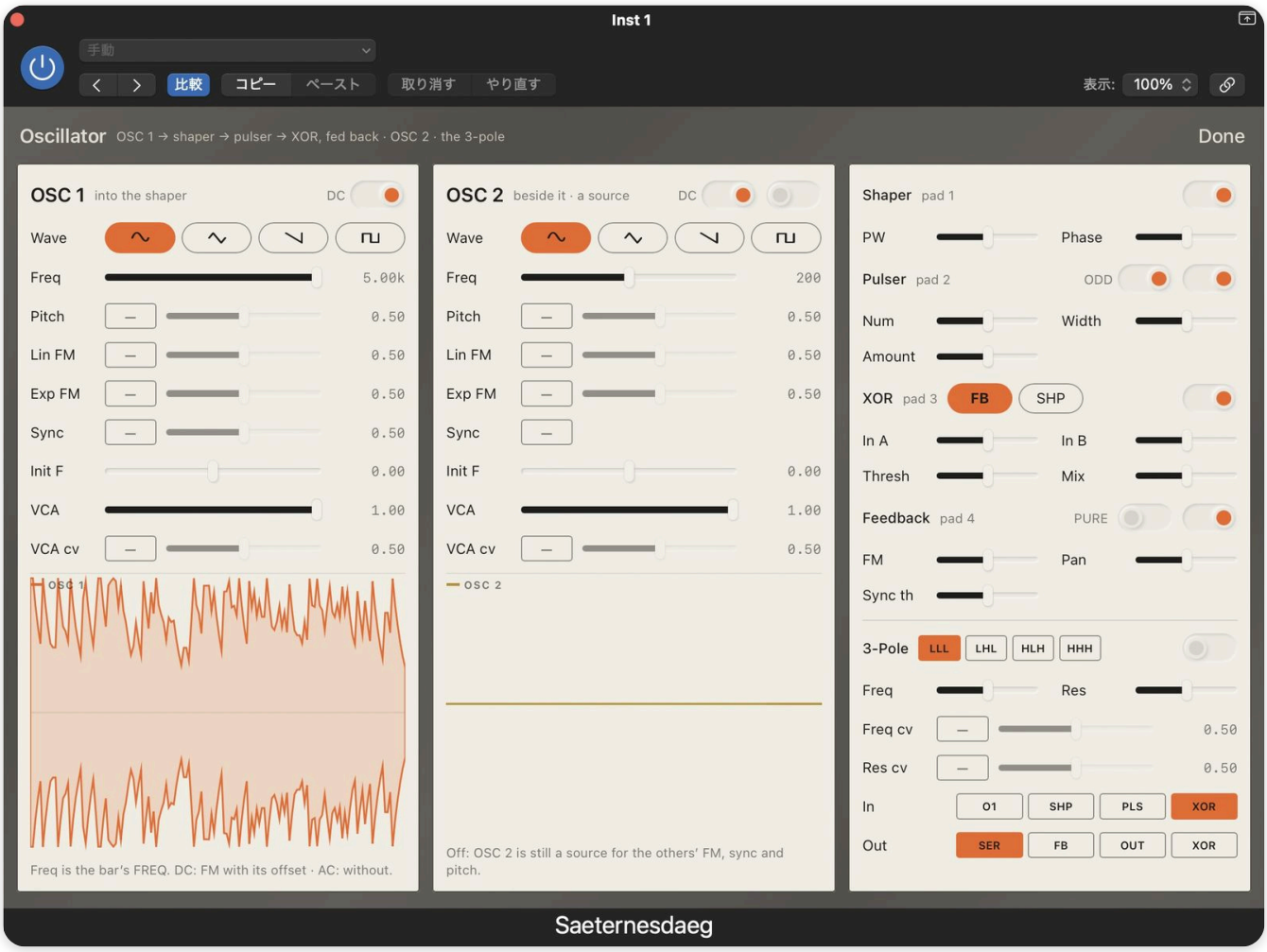
| | |
|---|---|
| FOLDER | The shaper: PW (X) and PHASE (Y). |
| PULSES | The pulser: NUM, how many pulses (X), and WIDTH (Y). |
| XOR | The XOR's two inputs: IN A (X) and IN B (Y). |
| FM / PAN | The feedback: FM (X) and PAN (Y). With PURE on (Oscillator panel), X is the volume: VOL / PAN. |
| The cross | The dotted sliders between the pads, each half its own: left, the XOR's threshold · right, the XOR mix (the pulser dry, or the XOR) · up, the sync threshold · down, the pulser amount. |

## The moon and the camera

The round key where the pads meet: the camera. On, the four pads show the picture and their pointers are drawn toward what moves; the pads turn grey and take no hand while it is on. Tap again: off.

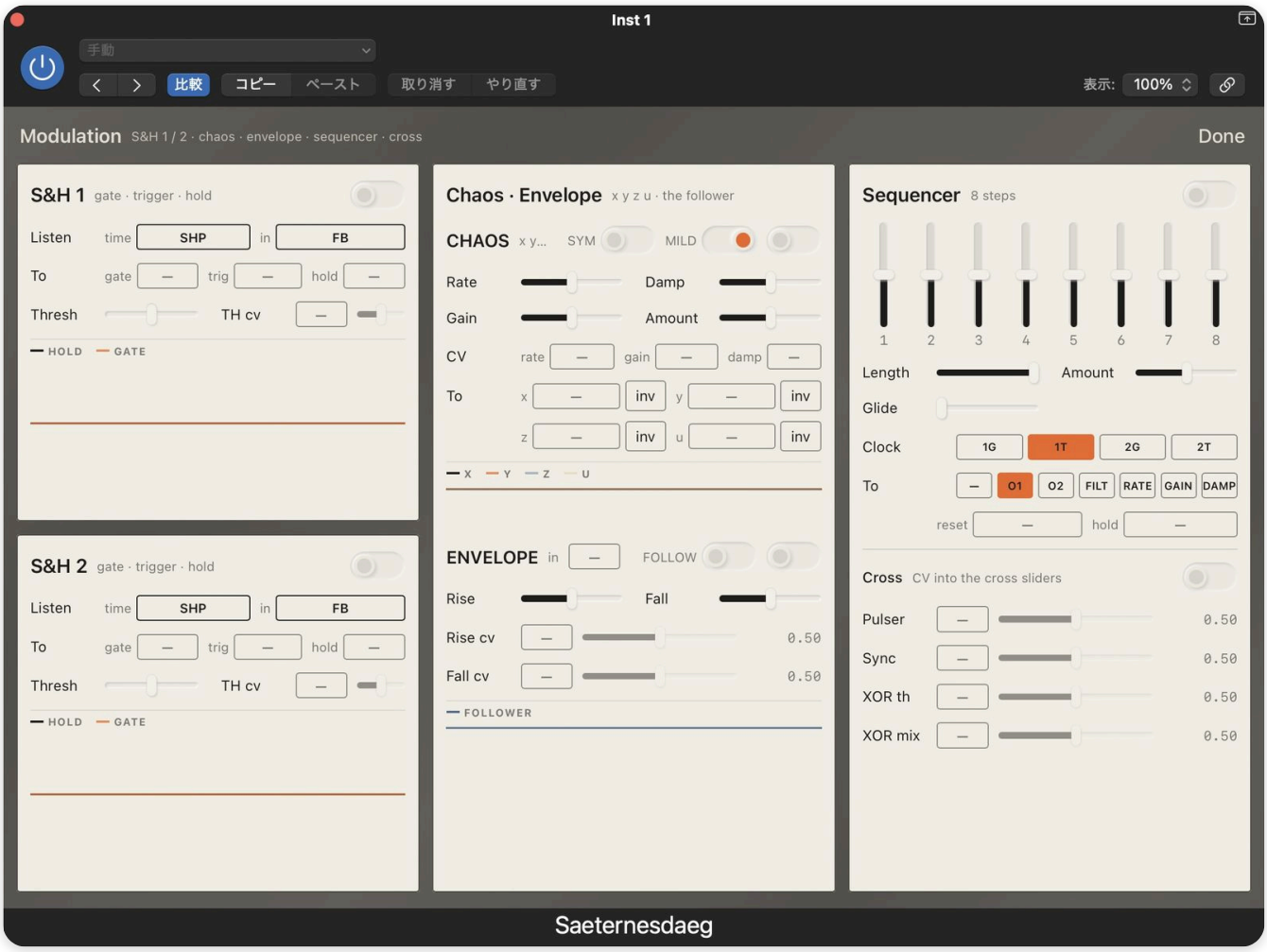


# Oscillator



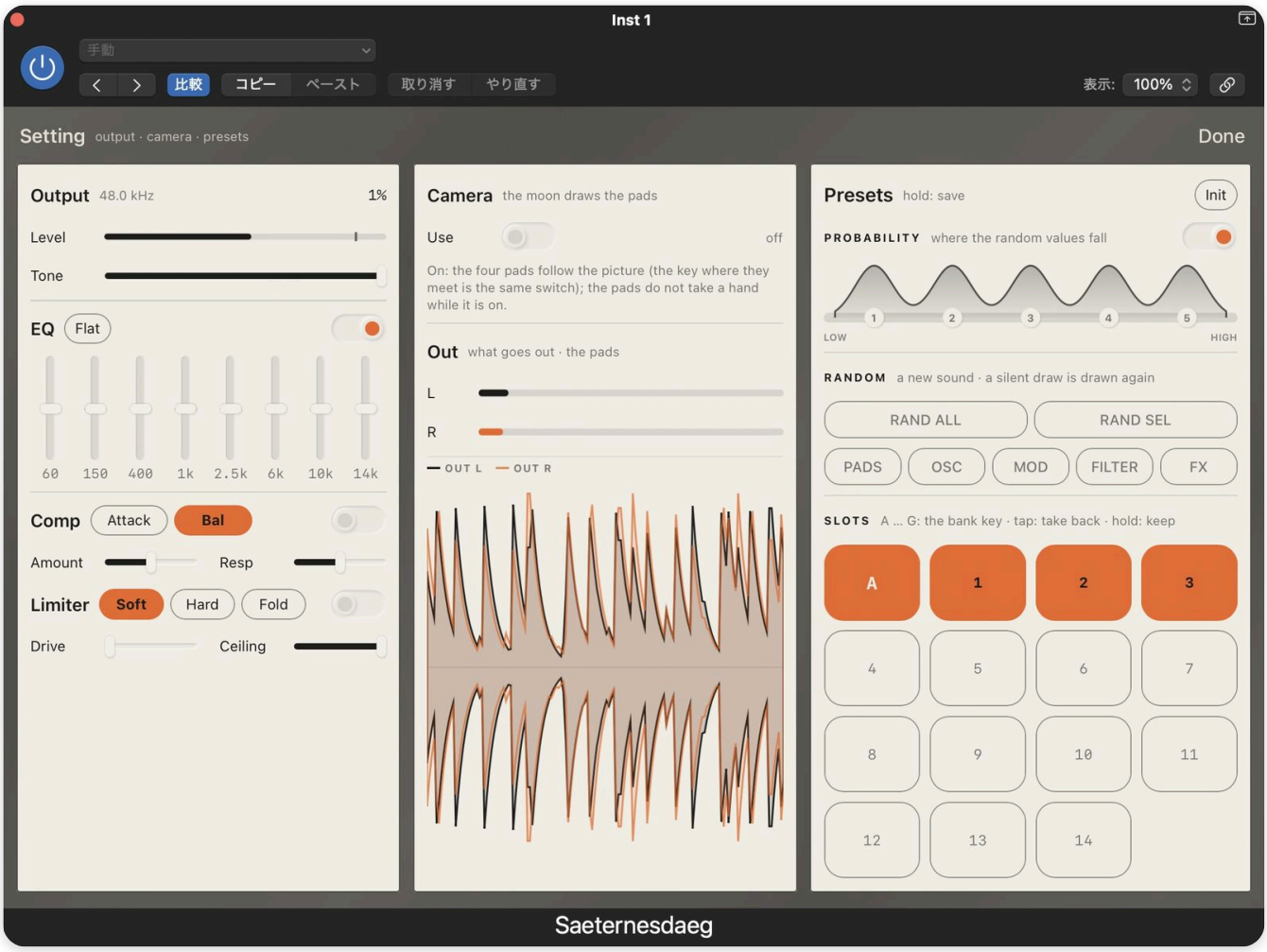

| | |
|---|---|
| OSC 1 | Into the shaper. WAVE, FREQ, and what moves it: PITCH, LIN FM, EXP FM, SYNC (each a source and an amount), INIT F, VCA, VCA cv. DC / AC: FM with or without its offset. |
| OSC 2 | Beside it, a source for the others. The same rows; its switch turns its sound on or off. |
| Shaper · Pulser · XOR · Feedback | The four pads' knobs, and their switches. ODD / EVEN: the pulser's pulses. B FROM: FB or SHP. PURE: pad 4's X is the volume. |
| 3-Pole | LLL · LHL · HLH · HHH, Freq, Res, their CV, **In** (where it listens) and **Out** (SER in series · FB into the feedback · OUT to the output · XOR into the XOR's sum). |

# Modulation




| | |
|---|---|
| S&H 1 · 2 | Gate, trigger, hold. Listen: what times it and what it holds. To: where its gate, trigger and hold go. Thresh, TH cv. Its scope: the held value and the gate. |
| Chaos · Envelope | CHAOS: Rate, Damp, Gain, Amount, SYM / ASYM, MILD / WILD, their CV, and x y z u to four places (inv: upside down). ENVELOPE: a follower or, with CYCLE, a slope that runs by itself. Each with its scope. |
| Sequencer · Cross | Eight steps, Length, Amount, Glide, its Clock and where it goes (To), reset and hold. Cross: CV into the four cross sliders. |

# Setting




| | |
|---|---|
| Output | Level (the meter), Tone, an eight-band EQ (**Flat**), Comp (Attack / Bal, Amount, Resp) and Limiter (Soft / Hard / Fold, Drive, Ceiling). |
| Camera · Out | Use: the camera on or off — the same switch as the moon. Out: what goes out, L and R. |
| Presets | Init, PROBABILITY, RANDOM and the SLOTS — see Presets and random. |

# Eight voices

The circuit's parts as a playable synthesizer, in the order the sound goes — the arrows between the sections show the way. Touch a knob: its name and value in its section's head. Double-click: its first value. The preset row above the face (A … G, 1 … 14) is SYNTH's own.




### How it is built

Each voice is ORIGIN's circuit taken apart and laid in a line, played from keys: the same oscillators, shaper, pulser, XOR and 3-pole filter, with an envelope and a gate after them instead of the feedback loop.

TZ OSC 1 ⇄ 2 → SHAPER → PULSER → XOR → MIXER · 3-POLE → GATE → out

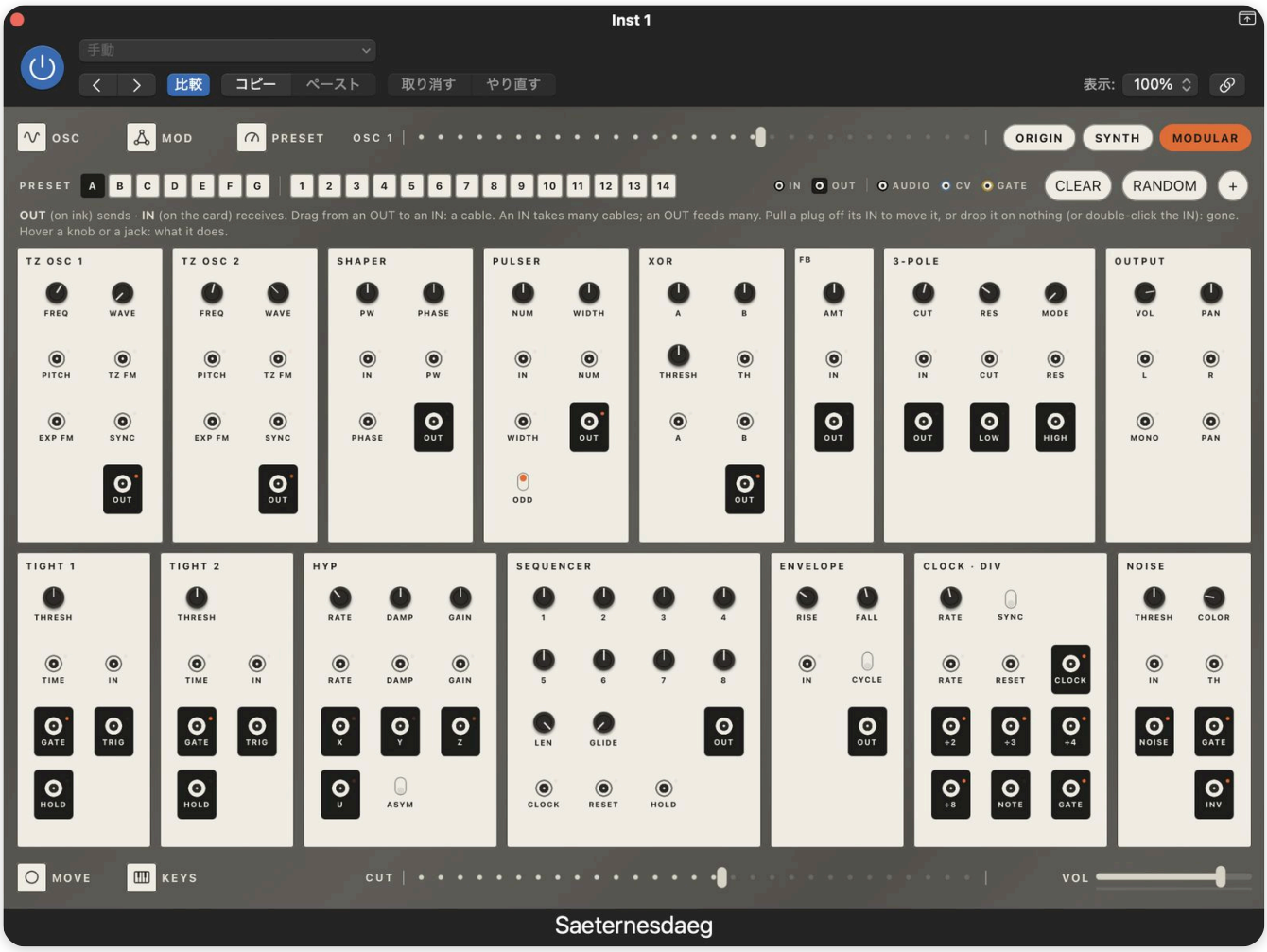
The two oscillators modulate each other first (FM 2→1, FM 1→2); their sum, at the MIXER's OSC 1 and OSC 2 levels, runs through the shaper, the pulser and the XOR, then the 3-pole filter and the gate. The XOR's B input listens to OSC 2 or to the shaper.

| | |
|---|---|
| 01 · TZ OSC 1 ⇄ 2 | Two oscillators that can run through zero, each modulating the other. **POLY** (eight voices) or **MONO** (one, legato, with GLIDE). WAVE 1, WAVE 2, TUNE, RATIO (OSC 2 against OSC 1), FM 2→1 and FM 1→2 (each way), FINE. **SNAP** keeps RATIO on harmonics and half harmonics — the bells. **SYNC** restarts OSC 1 with OSC 2. |
| 02 · SHAPER | AMOUNT, PW, PHASE: the wave folded and clipped. |
| 03 · PULSER | AMOUNT, NUM, WIDTH: pulses cut from the wave. |
| 04 · XOR | A window over the sum of A and B. B from OSC 2 or the SHAPER. MIX, THRESH. |
| 05 · MIXER · 3-POLE | OSC 1 and OSC 2's levels, then the 3-pole filter: LLL · LHL · HLH · HHH, CUT, RES, ENV (the filter envelope, ±), KEY (follows the note). Its curve is drawn beneath. |
| 06 · GATE | VCA (the level) · LP (a low pass that closes) · BOTH. LEVEL, and VELO: how much the velocity counts. |
| 07 · ENVELOPES | FILTER and AMP, each A · D · S · R. |
| 08 · HYP | A chaotic modulator: RATE, DAMP, GAIN. Its four outputs x y z u each go to one knob: **TO** opens a list (a section, then its knob), AMT sets how much, ±. |
| RANDOM | A new sound: pluck, pad, bass, lead, drone or bell, and around it. Its kind is shown beside it. |

**What moves what**

| | |
|---|---|
| FILTER ENV | Opens and closes the 3-pole filter, by ENV (±). KEY makes the cutoff follow the note. |
| AMP ENV | The gate: its level (VCA), its low pass (LP), or both. VELO: how much a key's velocity counts. |
| HYP | One chaotic modulator shared by every voice; its x y z u each move one knob of the face. |
| POLY | Eight voices. A new key takes a free voice; the same key again starts its voice from where it is; a ninth key takes the oldest voice, faded out first — no clicks. |
| MONO | One voice, legato: a new key while one is held only glides there (GLIDE); let go and it returns to the key still held. |

# Only the cables




Every inner connection is cut: nothing sounds until a cable reaches **OUTPUT**. **OUT** jacks sit on ink plates, **IN** jacks on the card; the ring's colour says what it carries — **AUDIO** (orange), **CV** (blue) or **GATE** (amber).

### How the case is laid out

The top row is the sound path, in the order it usually runs: TZ OSC 1 and 2, SHAPER, PULSER, XOR, FB, 3-POLE and OUTPUT. The bottom row moves it: TIGHT 1 and 2 (sample & holds), HYP (chaos), SEQUENCER, ENVELOPE, CLOCK · DIV and NOISE. Any OUT may go into any IN; the ring's colour only says what is usual. **RANDOM** picks one of several circuits for the top row, adds three to seven modulators from the bottom row, and keeps the patch only if it is heard on both sides.

**A first patch**

| | |
|---|---|
| 1 | TZ OSC 1 **OUT** → SHAPER **IN**, SHAPER **OUT** → 3-POLE **IN**. |
| 2 | 3-POLE **OUT** → OUTPUT **L** and **R**. Now it sounds. |
| 3 | HYP **X** → 3-POLE **CUT**: the filter moves by itself. |
| 4 | CLOCK · DIV **CLOCK** → TIGHT 1 **TIME**, HYP **Y** → TIGHT 1 **IN**, TIGHT 1 **HOLD** → TZ OSC 1 **PITCH**: a stepped melody. |

| | |
|---|---|
| Cables | Drag from an OUT to an IN. An IN takes many cables (they add); an OUT feeds many. Pull a plug off its IN to move it, or drop it on nothing — or double-click the IN — to remove it. |
| CLEAR RANDOM + | **CLEAR**: every cable off. **RANDOM**: a new patch, one of several circuits with modulators — always heard on both sides. **+**: two more cables onto the patch there is. |
| The line | Under the bar: what the jack or the knob under the pointer does. |

## The modules

| | |
|---|---|
| TZ OSC 1 · 2 | FREQ, WAVE. IN: PITCH, TZ FM (through zero), EXP FM, SYNC. OUT. |
| SHAPER | PW, PHASE, each with its IN. IN, OUT. |
| PULSER | NUM, WIDTH, each with its IN. ODD. IN, OUT. |
| XOR | A, B (their levels), THRESH and its IN TH. IN A, IN B. OUT. |
| FB | AMT: the feedback, IN to OUT. |
| 3-POLE | CUT, RES, MODE. IN, CUT, RES. OUT (its mode), LOW, HIGH. |
| OUTPUT | VOL, PAN. L, R, MONO (panned by PAN and its IN). |
| TIGHT 1 · 2 | Two sample & holds: THRESH. IN, TIME. OUT: GATE, TRIG, HOLD (the value held). |
| HYP | Chaos: RATE, DAMP, GAIN, each with an IN. ASYM. OUT: X, Y, Z, U. |
| SEQUENCER | Eight steps (1 … 8), LEN, GLIDE. IN: CLOCK, RESET, HOLD. OUT. |
| ENVELOPE | A follower: RISE, FALL, IN. CYCLE: it rises and falls by itself. OUT. |
| CLOCK · DIV | RATE, SYNC (to the DAW). IN: RATE, RESET. OUT: CLOCK, ÷2, ÷3, ÷4, ÷8, and the keyboard's NOTE and GATE. |
| NOISE | THRESH, COLOR. IN, TH. OUT: NOISE, GATE (the comparator), INV. |

The top row is the sound path; the bottom row moves it. The dot by an OUT shows what it does now.

# Keep it, draw it anew

| | |
|---|---|
| A 1 2 3 … | Each mode has its own presets: seven banks (A … G) of fourteen. In Setting, the first tile is the bank key: tap for the next. In SYNTH and MODULAR the same row sits above the face. |
| Tap · hold · ⌥ | Tap a number: the sound kept there comes back. Hold: keep what you hear there (it blinks). ⌥-click: clear it. A lit number holds a sound. |
| Init | The first sound of this mode. |
| RAND ALL<br>RAND SEL | **RAND ALL**: a new sound, every knob drawn anew. **RAND SEL**: a few of them. A draw that would be silent is drawn again. |
| PADS OSC<br>MOD FILTER<br>FX | ORIGIN: only that part drawn anew. |
| PROBABILITY | Where the random values fall. Five points over LOW … HIGH, each a hump: drag a point, and the draws land under the curve. Its switch off: even, anywhere. |

### Tips

| | |
|---|---|
| A drone | In ORIGIN keep a key held — or none: the oscillator goes on sounding. MOVE records the pads moving, to loop under it. |
| Bells | SYNTH: sine waves, SNAP on, a ratio such as ×3.5, some FM 2→1, a short AMP envelope. |
| A patch to start from | MODULAR: RANDOM, then + a few times; pull plugs off to hear what each cable did. |
| In a DAW | Automate the pads (Pad 1 X … Pad 4 Y) and Volume; MODULAR's clock follows the DAW with SYNC on. |

### Credits

Sæternesdæg — k.odk (Kei Odaka) · instagram.com/k_odk_

Requires macOS 12 or later · Apple silicon and Intel. Questions: instagram.com/k_odk_ · kodk.gumroad.com